# Band-1 Computational Experiment: van der Waerden Numbers — Finite Ledger + Box Certificates for W(2,3)=9 and W(2,4)=35

Human + ZoraASI full-dash (C.M. Baird / Band-1 dash worker)

2026-09-11

## Band-1: van der Waerden Numbers — Small Exact Status

**Honesty stamp.** This note does **not** claim that the van der Waerden problem (asymptotics / growth of $W(r,k)$) is solved. It does **not** claim any Clay or ToE stamp. **GATE-04 collection remains LOCKED** (out of scope). We record a literature ledger of known exact values and independently box-certify two classical small exact values: $W(2,3)=9$ and $W(2,4)=35$. Full asymptotic theory remains **OPEN**.

### Abstract

Van der Waerden's theorem guarantees that for every $r,k\geq 1$ there exists a least $N=W(r,k)$ such that every $r$-colouring of $\{1,\ldots,N\}$ contains a monochromatic arithmetic progression of length $k$. Only finitely many nontrivial exact values are known. Tonight's Band-1 probe:

1. Compiles a careful literature ledger (OEIS A005346 / A171081, Chvátal, Stevens–Shantaram, Kouril–Paul, Kouril, Wikipedia/MathWorld survey cutoff 2026-09).
2. Independently certifies on this Linux box, by pruned backtracking:
   - $W(2,3)=9$ (free 2-colourings of [8]: **6**; of [9]: **0**);
   - $W(2,4)=35$ (explicit $\mathrm{AP}_4$-free colouring of [34]; **0** free colourings of [35]).
3. Names the **next open diagonal exact targets**: $W(2,7)$ (lower $>3703$), $W(3,5)$ (lower $>2173$), $W(4,4)$ (lower $>1048$).

No new records are claimed.

### Notation

| Symbol | Meaning |
|---|---|
| $W(r,k)$ | classical van der Waerden number ($r$ colours, mono AP length $k$) |
| $w(r;k_1,\ldots,k_r)$ | off-diagonal: colour $i$ forces AP of length $k_i$ |
| OEIS A005346 | sequence $W(2,n)$ for $n=1,2,\ldots$ |
| OEIS A171081 | off-diagonal $w(2;3,n)$ |

Equivalences used below: $w(2;3,3)=W(2,3)=9$, $w(2;4,4)=W(2,4)=35$. (The informal tags `$W(3,3)=9$'' /`$W(3,4)=35$" in some dash notes mean these 2-colour diagonal cases, not $W(3,3)=27$ / $W(3,4)=293$.)

### Literature ledger (not re-proved except the two box certificates)

**Exact diagonal values (published)**

| $W(r,k)$ | Value | Citation |
|---|---|---|
| $W(1,k)$ | $k$ | trivial |
| $W(r,2)$ | $r+1$ | trivial |
| $W(2,3)$ | **9** | Chvátal 1970; **box-certified tonight** |
| $W(2,4)$ | **35** | Chvátal 1970; **box-certified tonight** |
| $W(2,5)$ | 178 | Stevens–Shantaram 1978; OEIS A005346 |
| $W(2,6)$ | 1132 | Kouril–Paul, *Exp. Math.* 17 (2008) |
| $W(3,3)$ | 27 | Chvátal 1970 |
| $W(3,4)$ | 293 | Kouril, *Integers* 12 (2012) A46 |
| $W(4,3)$ | 76 | Beeler–O'Neil 1979 |

**Next open diagonal exact values (Band-1 frontier)**

| Target | Status | Best cited lower (survey) |
|---|---|---|
| $W(2,7)$ | **OPEN exact** | $> 3703$ |
| $W(3,5)$ | **OPEN exact** | $> 2173$ |
| $W(4,4)$ | **OPEN exact** | $> 1048$ |
| $W(5,3)$ | OPEN exact | $> 170$ |

**Selected off-diagonal exact values**

| $w(\ldots)$ | Value | Citation |
|---|---|---|
| $w(2;3,4)$ | 18 | Chvátal 1970 |
| $w(2;3,5)$ | 22 | Chvátal 1970 |
| $w(2;3,6)$ | 32 | Chvátal 1970 |
| $w(2;3,7)$ | 46 | Chvátal 1970 |
| $w(2;3,20)$ | 389 | Kouril (verified) |
| $w(2;4,5)$ | 55 | Chvátal 1970 |
| $w(3;3,3,4)$ | 51 | Beeler–O'Neil 1979 |

Sources: Wikipedia *Van der Waerden number*; MathWorld; OEIS A005346, A171081; Landman–Robertson–Culver; Ahmed–Kullmann–Snevily.

**Asymptotics (explicitly OPEN)**

Shelah: $W(r,k)$ is primitive recursive. Gowers: enormous tower-type upper bounds from quantitative Szemerédi. Lower bounds of Berlekamp / later SAT constructions remain far below. **No asymptotic claim is made in this probe.**

## Method (box)

Represent a 2-colouring of $[n] = \{1, \ldots, n\}$ as a length-$n$ list over $\{0, 1\}$ (printed as `R/B`). After colouring integer $m$, reject if any monochromatic $k$-term AP **ends at** $m$:

$$a,\ a+d,\ \ldots,\ a+(k-1)d = m \quad \text{all the same colour.}$$

1. **Lower bound $W(2,k) > n$:** find one surviving colouring of $[n]$.
2. **Upper bound $W(2,k) \le n+1$:** exhaustive pruned DFS shows no survivor on $[n+1]$.
3. **Census:** count survivors for small $n$ (methods sanity).

Implementation: pure Python 3 (`verify_vdw.py`); no external SAT solver required at these sizes.

## Results (numbers!)

**Certificate A** — $W(2,3) = 9$

| Check | Outcome |
|---|---|
| Witness on [8] | `RRBBRRBB` |
| Mono 3-AP in witness? | **none** |
| Free 2-colourings of [8] | **6** |
| Free 2-colourings of [9] | **0** |
| Upper-bound DFS wall time | $\approx 0.00014\,\mathrm{s}$ |
| Conclusion | $W(2,3) = 9$ (independent re-verification) |

**Certificate B** — $W(2,4) = 35$

| Check | Outcome |
|---|---|
| Witness on [34] | `RRBRRRBBBRBRRBRRRBBBRBRRBRRRBBBRBB` |
| Mono 4-AP in witness? | **none** |
| Lower DFS | $\approx 0.0115\,\mathrm{s}$ / 1769 nodes |
| Free colourings of [35] | **0** |
| Upper DFS | $\approx 0.127\,\mathrm{s}$ / 20351 nodes |
| Conclusion | $W(2,4) = 35$ (independent re-verification) |

**Small census (2-colourings free of mono 3-AP)**

| $n$ | $AP_3$-free | $AP_4$-free |
|---|---|---|
| 1 | 2 | 2 |
| 2 | 4 | 4 |
| 3 | 6 | 8 |
| 4 | 10 | 14 |
| 5 | 14 | 26 |
| 6 | 20 | 48 |
| 7 | 16 | 78 |
| 8 | **6** | 132 |
| 9 | **0** | 230 |
| 10 | 0 | 356 |
| 11 | 0 | 548 |
| 12 | 0 | 842 |

The drop to **0** at $n = 9$ for $AP_3$ is exactly the upper half of $W(2,3) = 9$.

## What was not done

- No claim on $W(2,5) = 178$, $W(2,6) = 1132$, $W(3,3) = 27$, $W(3,4) = 293$ beyond literature citation.
- No search for $W(2,7)$ / $W(3,5)$ / $W(4,4)$ (open; heavy SAT/cluster territory).
- No asymptotic improvement; no Szemerédi-density claim.
- **GATE-04 collection: LOCKED.**

## Reproducibility

```
cd /workspace/mission-control/experiments/band1/vandervaerden_finite
python3 verify_vdw.py
```

Artifacts: `verify_vdw.py`, `results.json`, `literature_ledger.json`, `SUMMARY.txt`. Mirror: `/workspace/experiments/ban`

## Conclusion

**Band-1 deliverable:** honest finite ledger + box certificates for the two smallest nontrivial classical values $W(2,3) = 9$ and $W(2,4) = 35$, with the next open diagonal exact targets named. Prefer exact verification of known small $W$ over claiming new records. Asymptotic van der Waerden growth remains **OPEN**.